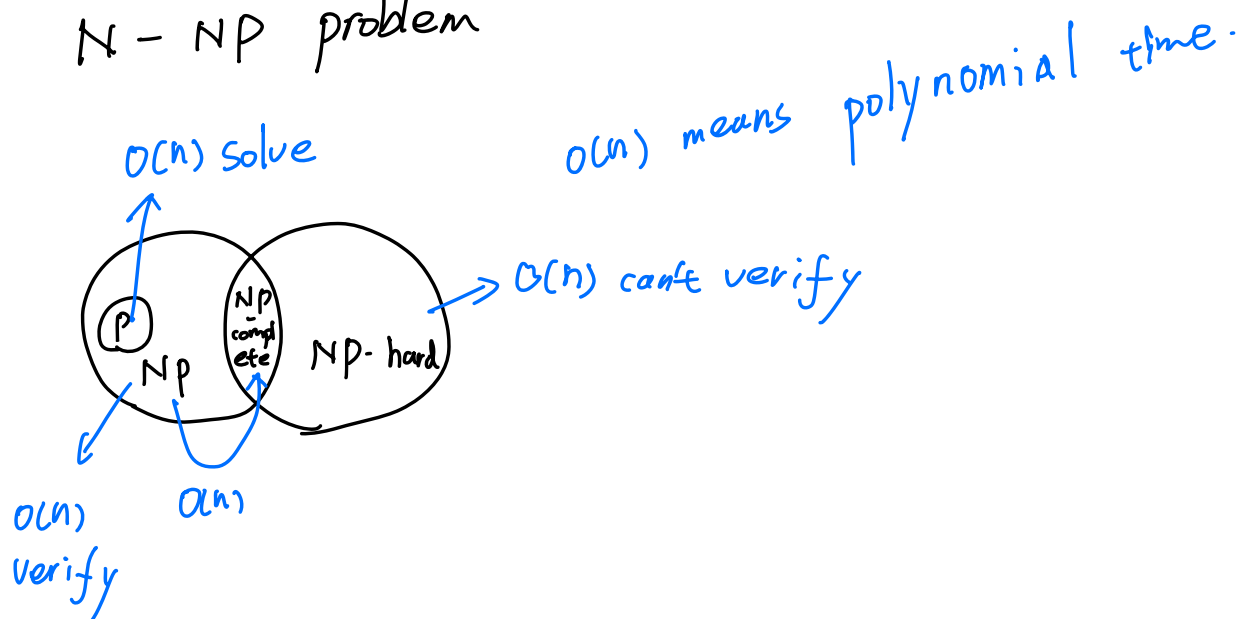


输入-输出 { 建模 modeling $\checkmark \rightarrow \boxed{?} \rightarrow \checkmark$
 优化 optimisation $? \rightarrow \boxed{\checkmark} \rightarrow \checkmark$
 模拟 simulation $\checkmark \rightarrow \boxed{\checkmark} \rightarrow ?$

Optimization vs constraint

opt. con	$\checkmark$	$\times$
$\checkmark$	COP	CSP
$\times$	FOP	No problem

N - NP problem



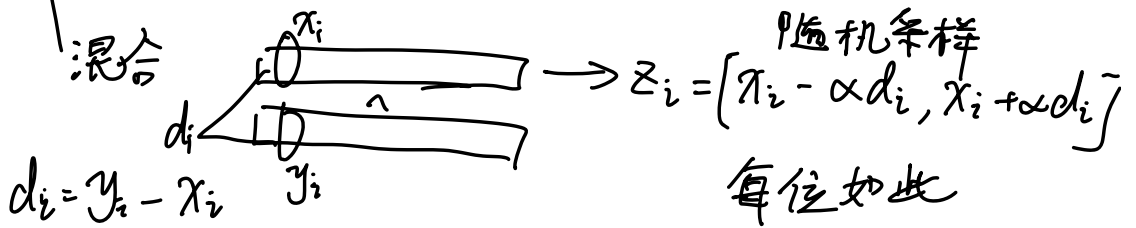
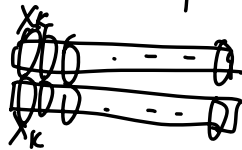
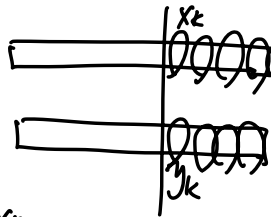
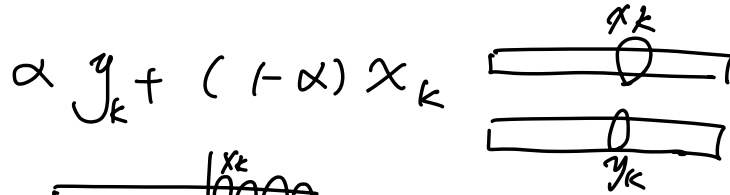
交叉

二进制

- 一点
- n点
- 均匀

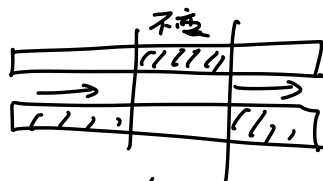
实数

单
简单
全
混合

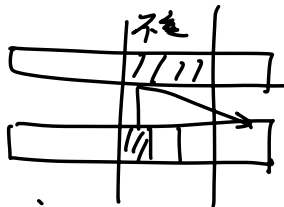


排列

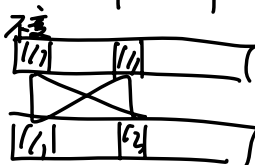
顺序



PMX



circle



边缘重组

choice	element select	Reason	Partial result
		+ → 子代	

奇变

遗传算法

$$2 \rightarrow 10 \quad x + \frac{y-x}{2^L-1} \times \sum_{j=0}^{L-1} a_{L-j} 2^j$$

eg. 4 bit $\rightarrow [-10, 10]$
 $\begin{matrix} L & & x & y \end{matrix}$

$\langle a_1 a_2 a_3 a_4 \rangle$ $\begin{matrix} 1111 \\ 8421 \end{matrix} \rightarrow 15$

$$-10 + \frac{20}{15} \times 0 = -10$$

$$-10 + \frac{20}{15} \times 1 = -8.6667$$

$$-10 + \frac{20}{15} \times 15 = 10$$

$$10 \rightarrow 2 \quad B = \text{round} \left[\frac{z-x}{y-x} (2^L-1) \right]$$

eg. $[-10, 10] \rightarrow 4 \text{ bit}$
 $\begin{matrix} x & y \end{matrix}$

$$B = \text{round} \left(\frac{z+10}{20} \times 15 \right)$$

$$z = -10, B = 0 \quad 0000$$

$$z = -9, B = 1 \quad 0001$$

$$z = 0, B = 8 \quad 1000$$

2进制 Disadv.

- ① 只有 2^L 个有限值
- ② L 决定了可能达到的最大精度
- ③ 高精度意味着长的染色体 chromosomes, 评估计算量大, 速度慢.

SGA 步骤

- ① 编码 Encoding
- ② 初始种群 initiate
- ③ 解码 Decoding
- ④ 计算适应度比例: roulette wheel
- ⑤ 交叉 cross over
- ⑥ 变异 mutation

数据类型

Qualitative < Nominal
Ordinal Quantitative < Interval
Ratio

missing value
 remove
 Imputation: mean, median, mode
 estimate

异常值
outlier
 remove ~
 Imputation: mean, median, mode
 capping : 5th percentile
 95th percentile
 treated separately
 shouldn't amend it

Standardisation

$$x' = \frac{x - \mu}{\sigma}$$

$\mu=0$ $\sigma=1$	正态	不敏感	留分布 改关系
0-1	非正态	敏感	改分布 留关系

normalization

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

(Min - Max scaling)

贝叶斯决策 ① 已知 m_1, m_2, C_1, C_2, x

$$\begin{aligned} \textcircled{2} g_i(x) &= p(x|w_i) p(w_i), \quad i=1,2 \\ &= \frac{1}{(2\pi)^{\frac{d}{2}} |C_i|^{\frac{1}{2}}} e^{-\frac{1}{2}(x_i - \mu_i)^T C_i^{-1} (x_i - \mu_i)} \cdot p(w_i) \end{aligned}$$

$$\textcircled{3} g_1(x) > g_2(x) \rightarrow \text{class 1}$$

$$g_1(x) < g_2(x) \rightarrow \text{class 2}$$

$$g_1(x) = g_2(x) \rightarrow x \text{ on the decision boundary (undecided)}$$

Naïve

$$\textcircled{1} \text{ 已知 } m_1, m_2, C_1, C_2, \text{ sample } x = [x_1, x_2]^T$$

$$\begin{aligned} \textcircled{2} \text{ 已知 Class 1: } &\mu_{11}, \mu_{21}, \sigma_{11}^2, \sigma_{21}^2 \\ \text{Class 2: } &\mu_{12}, \mu_{22}, \sigma_{12}^2, \sigma_{22}^2 \end{aligned}$$

$$\textcircled{3} g_i(x) = \prod_{j=1}^n p(x_j | w_j) p(w_j)$$

$$p(x_i | w_j) = \frac{1}{\sqrt{2\pi} \sigma_{ij}} e^{-\frac{1}{2} \left(\frac{x_i - \mu_{ij}}{\sigma_{ij}} \right)^2}$$

$$g_1(x) = p(x_1 | w_1) p(x_2 | w_1) p(w_1)$$

$$g_2(x) = p(x_1 | w_2) p(x_2 | w_2) p(w_2)$$

$$\textcircled{4} g_1(x) > g_2(x) \rightarrow \text{class 1}$$

$$g_1(x) < g_2(x) \rightarrow \text{class 2}$$

$$g_1(x) = g_2(x) \rightarrow x \text{ on the decision boundary (undecided)}$$

Fisher linear discriminant

① 已知 m_1 m_2 S_1 S_2 $y = w^T x + w_0$ x_2

② $S_w = S_1 + S_2$ $C_i = \frac{1}{n} S_i$ $\Sigma_i = \frac{1}{n} S_i$

$$w = S_w^{-1} (m_1 - m_2)$$

$$w_0 = - \frac{w^T (m_1 + m_2)}{2}$$

SVM

① 已知支持向量 x_1 x_2 x_3 x_4 , $N_s = 4$

Lagrange multipliers α_1 α_2 α_3 α_4

Label $d(i)$ 1 1 -1 -1

$$y = w^T x + w_0 \quad x_3$$

② $w^* = \sum_{i=1}^N \alpha(i) d(i) x(i)$

$$b^* = \frac{1}{N_s} \sum_{\alpha(i) > 0} [d(i) - (w^*)^T x(i)]$$

③ $y = w^T x + w_0 > 0 \rightarrow \text{class } 1$

$y = w^T x + w_0 < 0 \rightarrow \text{class } -1$

	LDA	SVM
Objective	$\max S_B, \min S_w$	$\max \text{margin}, \min \frac{1}{2} w^T w$
Data Usage	entire training data	support vector
Data Distribution	Gaussian distribution + same covariance matrix	don't make assumptions

SVM ① separable $\min \frac{1}{2} w^T w$

subject to $d(i) \times [w^T x(i) + b] \geq 1$

② non-separable $\min \left(\frac{1}{2} w^T w + c \sum_{i=1}^N \epsilon(i) \right)$

subject to $d(i) \times [w^T x(i) + b] \geq 1 - \epsilon(i)$

classification tree

Question ask ① one or more ② set by designer

③ branch factor/ratio ④ binary/multiple splits

Attribute ① Information Gain ② Gain Ratio ③ Gini index

Leaf ① cross-validation ② threshold value in the reduction in impurity

pruning ① pre ② post

assignment ① single category ② labelled by most sample

impurity ① Entropy $i_E(N) = - \sum_j p(w_j) \log_2 p(w_j)$

② Gini $i_{\text{Gini}}(N) = \sum_{j \neq k} p(w_j) p(w_k)$
 $= \frac{1}{2} \left[1 - \sum_j p^2(w_j) \right]$

Random forest ① Bootstrapping: 有放回重采样

② Aggregating: majority voting 聚合

Regression ① Multiple regression

more than one independent variables

② Ordinary least square estimation

$$J(\theta) = \frac{1}{2} \sum_{i=1}^N \varepsilon_i^2 \quad \hat{\theta} = (\phi^T \phi)^{-1} \phi^T y$$

③ Ridge (L2) $J(\theta) = \frac{1}{2} \left\{ \sum_{i=1}^N (\hat{y}_i - y_i)^2 + \lambda \sum_{j=0}^m \theta_j^2 \right\}$

$$\hat{\theta} = (\phi^T \phi + \lambda I)^{-1} \phi^T y$$

④ Lasso (L1) $J(\theta) = \frac{1}{2} \left\{ \sum_{i=1}^N (\hat{y}_i - y_i)^2 + \lambda \sum_{j=0}^m |\theta_j| \right\}$

Model Evaluation ① Mean Squared Error (MSE)

$$MSE = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

② Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}$$

③ Mean Absolute Error (MAE)

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|$$

④ R-squared (R^2)

$$R^2 = 1 - \frac{RSS}{TSS} \quad \begin{array}{l} \text{— sum of squares of residual} \\ \text{— total sum of squares} \end{array}$$

$$RSS = \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad TSS = \sum_{i=1}^N (\bar{y} - y_i)^2$$

Classifier Methods ① Hold-out, repeat Hold-out

② k-fold cross validation, repeated ~

③ Leave-one-out

Metrics ① Accuracy, error rate

TP	FP
FN	TN

TP	FP
FN	TN

	Actual	
predict	TP	FP
	FN	TN

② Sensitivity, specificity

TP	FP
FN	TN

TP	FP
FN	TN

③ Precision, recall, F-score = $\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

TP	FP
FN	TN

TP	FP
FN	TN

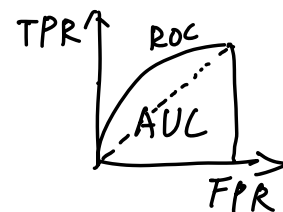
④ ROC, AUC

True positive Rate

TP	FP
FN	TN

False positive Rate

TP	FP
FN	TN



Feature subset selection

curse of dimensionality ① Data sparsity : not enough data

② Distance concentration

all the pairwise distance may converge to the same value

peaking phenomenon : ① feature $\uparrow$ initial improvement $\uparrow$ performance

② critical value after

feature $\uparrow$ probability of error $\uparrow$

Feature selection ① selecting a subset of feature out of original feature

Feature extraction : ① extracting or deriving information from the original features set to create a new feature subspace

② compress data + maintaining most relevant information

③ PCA

Individual

① 单独查特征, 测判别能力

② 非最优方法, 含不良特征

③ 类可分性好, 特征相关性差

Subset

① 选子集, 分类最佳

② 最大相关性, 最小冗余性

特征2部分

① 搜索 $\begin{cases} \text{穷举 exhaustive} \\ \text{前向 forward} \\ \text{后向 backward} \end{cases}$

② 评估 $\begin{cases} \text{分类器性能} \\ \text{可分性度量 separability} \end{cases}$

3算法

① 过滤 : classifier $\times$ separability $\checkmark$
filter

② 包装 : classifier $\checkmark$
wrapper

③ 嵌入式 : features penalize $\downarrow$
embedded

filter	wrapper	embedded
without training $\checkmark$ independence $\checkmark$ less overfitting $\checkmark$ scalability $\checkmark$	Direct optimization of model $\checkmark$ captures relationship $\checkmark$	balance efficiency & performance $\checkmark$ Automatically select feature $\checkmark$ reducing manual intervention Interpretability $\checkmark$
ignore feature interaction $\times$ misalignment goal $\times$ distribution assumption $\times$	repeated model training $\times$ overfitting risk $\times$	limited to models support $\times$ Hyperparameter sensitivity $\times$

Clustering

① Centroids-based / partitioning method : k-means

Data distribution uniform ✓
cluster boundaries clear ✓

① k ② centroid ③ closest ④ new ⑤ repeat ③④ → no change

② Hierarchical : Agglomerative

visualizing hierarchical data structures ✓
nested relationships ✓

① N clusters ② two closest, N-1 ③ N-2 ④ repeat 3 → one ⑤ dendrogram + divide

③ Density-based : DBSCAN

noise ✓ irregularly shape clusters ✓

① MinPts, ϵ , storing ② density connected → cluster ③ not visited → 2 ④ all + noise

④ Distribution-based : Gaussian mixture model

conforming to Gaussian distribution assumption ✓

probability interpretations ✓

① m ② $\hat{\alpha}_i(0)$ $\hat{\mu}_i(0)$ $\hat{\Sigma}_i(0)$ ③ EM ④ closest

metric 轮廓系数 (伦敦带卡来)

① silhouette coefficient

$$SC_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad a_i = \frac{1}{n_p - 1} \sum_{\substack{x_i, x_j \in D_p \\ i \neq j}} d(i, j) \quad b_i = \min_{\substack{x_j \in D_p \\ x_j \in D_q \\ i \neq j}} d(i, j) \quad SC = \frac{1}{N} \sum_{i=1}^N SC_i$$

平均距 类间最小

② Dunn index

$$DI = \frac{\min_{p \neq q} d_{pq}}{\max_{p \in K} \delta_p} \quad d_{pq} = \min_{\substack{x_i \in D_p \\ x_j \in D_q \\ p \neq q}} d(i, j) \quad \text{类间最小}$$

$$\delta_p = \max_{\substack{x_i, x_j \in D_p \\ i \neq j}} d(i, j) \quad \text{类内最大}$$

③ Davies - Bouldin index

$$DB = \frac{1}{K} \sum_{i=1}^K S_i \quad S_i = \max_{j \neq i} S_{ij} \quad S_{ij} = \frac{\delta_i + \delta_j}{d(i, j)} \quad \text{类内离散度}$$

$$\delta_i = \sqrt{\frac{1}{n_i} \sum_{x \in D_i} \|x - m_i\|^2} \quad m_i = \frac{1}{n_i} \sum_{x \in D_i} x \quad \text{双簇质心距}$$

④ Calinski - Harabasz index

$$CH = \frac{\sum_{i=1}^K n_i S_i}{\sum_{i=1}^K C_i} \quad C_i = \sum_{x \in D_i} \|x - m_i\|^2 \quad S_i = \|m_i - m\|^2$$

$$m = \frac{1}{N} \sum_{i=1}^N x_i$$

⑤ Rand index

$$R2 = \frac{Ns + Nd}{CC(N, 2)}$$